

Bulut Emre Sakarya

Software Engineer | AI & Automation · Backend · Mobile

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience building end-to-end products for the logistics and real estate industries, delivered under a SaaS model. I build autonomous AI agents (LangChain/LangGraph), high-scale web scraping pipelines, React Native mobile apps, and real-time BI dashboards. Two of my products are live on Google Play and the App Store, and every system I ship runs 24/7 on Linux VDS infrastructure I manage myself, with no dependency on no-code platforms. I design architectures that automate 90% of manual workflows, enforce SLAs automatically, and apply FinOps principles to LLM cost control — one system generated +1.5M TL in net revenue within 4 months.

TECHNICAL SKILLS

- Languages:** Python, JavaScript, SQL, HTML/CSS
- Backend & API:** FastAPI, Node.js, RESTful API design, webhook management, async queue architecture, OTP authentication
- AI & Orchestration:** LangChain / LangGraph (stateful multi-agent workflows), OpenAI API, structured data extraction (Pydantic/JSON Schema), prompt engineering, LLM cost optimization
- AI-Assisted Development:** Claude (Claude Code), ChatGPT, Gemini, Replit Agent — context management across large codebases, rapid prototyping, code review and refactoring; disciplined verification and testing of AI output before it ships
- Mobile:** React Native / Expo (EAS Build), push notifications, Google Play & App Store release pipelines, subscription / RTDN integration
- Data & BI:** PostgreSQL (PostGIS), SQLite (WAL Mode), Supabase, Drizzle ORM, ETL & data cleaning, dashboard development with React + TanStack Query
- Web Scraping:** Selenium, requests/aiohttp, anti-bot evasion, session & rate-limit management
- Cloud & DevOps:** AWS (Certified Cloud Practitioner, CLF-C02), Docker, Linux server administration (Ubuntu VDS, Oracle Cloud), PM2, Cloudflare Tunnel, Git & GitHub
- Languages (spoken):** Turkish (native), English (B2 — AWS certification exam completed in English), German (A1, beginner)

PROFESSIONAL EXPERIENCE

Freelance / Karaarslan Digital

Software Engineer (AI, Backend & Mobile)

11/2025 – Present

- Build AI agents, ETL pipelines, and mobile applications for SMEs and institutional clients, owning the full technical delivery from architecture design through store release.
- Tırdaş (02/2026 – Present):** Built an AI-powered logistics SaaS platform, WhatsApp dispatch bot, and mobile app, now live on Google Play and the App Store. Delivered the core build in 06/2026 and continue to run the platform under an ongoing paid maintenance, feature, and support retainer.
- Quadmate:** Shipped a location-based gaming-partner matchmaking app end to end (React Native + Supabase/PostGIS).
- Delivered corporate web, hosting, and SEO projects (CleanHome, Nilbaşak) and productized a resellable “Ready Solutions” line (AI customer agent, QR menu, KPI dashboard, data harvesting).
- Impact:** Automated 90% of clients' manual data entry operations, compressing an 8-hour weekly workload down to 5 seconds.

RE/MAX True

AI & Automation Programmer

09/2024 – 10/2025

- Joined as a real estate agent; internal automations I wrote to streamline field operations were noticed by management, and I moved into the AI & Automation Programmer role within two months (11/2024).
- Solely architected the “TrueMax” ecosystem — an anti-ban scraper, an SLA lead dispatcher, and an AI outreach orchestrator — and redeployed the same engine as RED-i for RedPartners.
- Expanded the office's active client portfolio by 60%+ and total branch revenue by 40%+, generating +1.5M TL in net revenue within 4 months.
- Designed and built the real-time BI dashboard tracking 40+ agents, cutting weekly reporting time from 8 hours to 5 seconds.

Özgelecek İK Teknolojileri A.Ş.

Data & Software Developer

04/2024 – 08/2024

- Developed Python automations for data processing and reporting, enabling the team to consistently exceed weekly output targets.
- Promoted from intern to Data & Software Developer within two months based on technical delivery.

Freelance

Python Developer

03/2020 – 01/2024

- Built custom backend automation scripts and web scrapers for e-commerce clients, and maintained the systems I delivered.

EDUCATION

Anadolu University

B.S. in Management Information Systems (Open Education)

2025 – Present

- Honor Certificate — Spring Semester, 2025–2026 Academic Year

Dokuz Eylül University

Undergraduate Studies in Physics (partial coursework)

2021 – 2023

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services (10/2025)
- Information System Design Certificate — Anadolu University (05/2026)
- JavaScript Course — Boğaziçi Enstitüsü (2026)
- IT Internship — TNC Group · Europäisches Bildungsinstitut (07/2025)
- BTK Akademi — AI Ethics, Customer Insight, Communication & Network Management (05/2025)

PROJECTS

Tırdaş — AI-Powered Logistics SaaS, Mobile App & WhatsApp Bot

Software Engineer (Solo)

An autonomous dispatching ops-center connecting logistics carriers with truck drivers. The moment a driver writes “I’m in Gebze, heading to Ankara, find me a load,” the system matches the best available freight, reports route weather, logs expenses, and escalates unresolved cases to a live support operator. The mobile app is live on Google Play and the App Store, sharing a single listing pool with the WhatsApp bot.

Python · FastAPI · LangGraph · PostgreSQL · React Native (Expo) · Meta Cloud API · PM2

- **Colloquial NLP Pipeline:** Built a LangGraph & JSON Schema workflow translating slang and misspelled trucking terminology (e.g., “Kırkayak” → 10-wheel truck) into strict, structured database queries.
- **Geographic Validation Layer:** Wrote a validation layer normalizing free-text listings against an 81-province / 973-district reference set; accompanying panel optimizations delivered a 300% speed-up.
- **FinOps Budget Engine:** Engineered an ApiManager module that calculates real-time token spend per user and throttles the system on reaching the monthly cost limit (monthly_cost_limit).
- **Mobile & Backend:** Delivered the React Native (Expo) client, FastAPI backend, OTP authentication, push notifications, admin panel, and Google Play subscription/RTDN integration end to end.
- **Async Message Queue:** Built a self-healing PM2 queue that ingests high-frequency Meta Webhook bursts without payload loss during server outages.

TrueMax.ai — AI Lead Generation & Outreach Pipeline

AI Automation Engineer (Solo)

A revenue engine that scrapes property listings, reaches owners with human-like WhatsApp outreach, and auto-distributes warm leads to agents. 30,000+ property owners reached, 2,778 collaboration approvals secured, across a scanned portfolio worth 39.1 billion TL. Generated +1.5M TL in net revenue for the office in 4 months.

Python · Selenium · Node.js · WhatsApp · LLM · SQLite

- **Anti-Ban Hybrid Scraper:** Designed a hybrid extraction engine capturing a real browser session (cookie/session) via Selenium, then offloading heavy traffic to fast requests.Session tunnels.
- **Human-Simulation Algorithm:** Bypassed bot detection with dynamic sendSeen() hooks and randomized 2–5 second jitter delays.
- **Strict 120-Minute SLA Engine:** Wrote the accountability module that assigns warm leads to agents and reassigns them Round-Robin to the next agent if uncontacted within 2 hours.
- **Cost Optimization:** Cut LLM spend by ~70% through prompt compression and caching, and redeployed the engine as RED-i for RedPartners.

Real-Time BI & Kiosk Dashboard

Frontend & Data Architect

A 24/7 lobby TV broadcast and management panel tracking real-time sales competition across 40+ agents, office revenue, and live currency/gold/crypto rates; ran in production on 215 real transactions.

React · Express · SQLite (WAL) · TanStack Query · Google Sheets API

- **Data Sanitization Layer:** Built RegEx pipelines normalizing manually entered malformed values (e.g., “1.500.00tl”, “1500000”), cutting the reporting error rate from 15% to 0.01%.
- **Lock-Free Concurrency:** Enabled SQLite WAL (Write-Ahead Logging) mode to prevent SQLITE_BUSY deadlocks during concurrent writes from two background bots.
- **Autonomous Kiosk Engine:** Coded a zero-refresh UI triggering the lobby browser via the DOM Fullscreen API and keeping data live with TanStack Query.

Quadmate — Location-Based Gaming Partner Matchmaking App

Software Engineer (Solo)

A mobile app helping players find the missing participant for a table or event nearby — users browse open tables on a map, join listings, or open their own table.

React Native (Expo) · Supabase · PostGIS · Push Notifications

- **Geospatial Matching:** Implemented radius-based PostGIS queries, row-level security (RLS), and real-time subscriptions on Supabase.
- **Mobile Delivery:** Shipped iOS and Android from a single codebase, owning the push notification pipeline and store release process.

ADDITIONAL DELIVERIES

- **Crypto Strategy Bot:** A scheduled strategy bot and control panel written using only the Python standard library — leverage, TP/SL, and position sizing controls with a live PnL feed; demoed to the client in paper-trading mode.
- **Telegram Ops Bot:** Deployed a notification bot on Oracle Cloud that monitors defined sources 24/7 and surfaces one-tap inline-keyboard actions when conditions are met.
- **Corporate Web & Hosting:** Delivered corporate sites for CleanHome and Nilbaşak with SSL, quote forms, SMTP/HTTP mail integration, and SEO foundations — running uninterrupted since delivery.

All interactive demos and technical case studies: bulutemresakarya.github.io